

Highlights

Threshold Selection as a Critical Design Choice in Financial Fraud Detection: A Systematic Comparison of Classical Machine Learning Models and Tabular Transformers

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- Fixed $\tau=0.5$ reduces F_2 to near zero on large-scale bank account fraud data
- The max- F_2 threshold strategy achieves up to $9\times F_2$ improvement over the fixed default
- SMOTE and SMOTE+Tomek Links produce identical results under extreme class imbalance
- Class Weights is the safest default imbalance strategy across model families
- A decision framework maps seven deployment scenarios to recommended configurations

Threshold Selection as a Critical Design Choice in Financial Fraud Detection: A Systematic Comparison of Classical Machine Learning Models and Tabular Transformers

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ABSTRACT

Financial fraud detection under extreme class imbalance is a well-studied problem, yet a recurring methodological shortcoming in the literature remains largely unaddressed: the decision threshold applied to model output scores is rarely treated as an explicit experimental variable. Most published studies either fix the threshold at $\tau = 0.5$ or select it on the test set, both of which are inappropriate practices that inflate reported metrics and prevent fair model comparison. This paper presents a systematic, leakage-free comparison of six model families — Logistic Regression, Random Forest, LightGBM, CatBoost, One-Class SVM, and FT-Transformer — under seven class imbalance handling strategies and four threshold selection strategies, evaluated on two structurally distinct fraud benchmarks: the ULB Credit Card Fraud dataset ($N = 284,807$, 0.17% fraud) and the Bank Account Fraud (BAF) Base dataset ($N = 1,000,000$, $\approx 1.1\%$ fraud). All threshold strategies are calibrated exclusively on inner validation folds and never touch the held-out test set, eliminating threshold leakage. On the ULB dataset, the fixed $\tau = 0.5$ default performs acceptably because model scores are well-separated; gradient-boosted models lead with CatBoost at PR-AUC = 0.885. On BAF, the fixed default is catastrophic: F_2 drops below 0.04 for every supervised model, while the max- F_2 strategy recovers F_2 values in the 0.29–0.32 range — an improvement of up to nine-fold. No single imbalance handling strategy universally dominates; Class Weights is the safest default. SMOTE and SMOTE+Tomek Links produce identical results under extreme imbalance (Tomek links remove zero borderline examples), a finding with practical significance for strategy selection. On the larger BAF dataset, the FT-Transformer matches CatBoost at PR-AUC = 0.180, demonstrating that tabular transformers are competitive but not superior to well-tuned gradient boosting. We provide a decision framework mapping seven deployment scenarios to recommended model–strategy–threshold configurations, directly usable by fraud detection practitioners.

1. Introduction

Financial fraud in electronic payment systems causes annual global losses in the hundreds of billions of dollars, and the proliferation of digital payment channels has expanded the attack surface available to fraudsters (Ryman-Tubb, Krause and Garn, 2018). Automated fraud detection systems, trained on historical transaction data, are now a critical component of financial institutions' defence infrastructure. The machine learning problem underlying fraud detection is characterised by three structural challenges: (i) extreme class imbalance, with fraud rates typically between 0.1% and 2% (He and Garcia, 2009); (ii) non-stationarity, as fraud patterns evolve in response to deployed detection systems; and (iii) asymmetric error costs, where missed fraud carries direct financial losses and excessive false positives burden investigation teams and degrade customer experience.

Despite the volume of published work, a systematic survey of the literature reveals recurring methodological limitations that undermine the reliability of reported conclusions. The most consequential is *data leakage*: applying resampling techniques before the train/test split, or without isolating resampling to the training partition of each cross-validation fold, inflates reported metrics by margins sufficient to reverse model rankings (Hayat and Magnier, 2025). A second limitation is the widespread use of ROC-AUC as the primary evaluation metric, which is systematically misleading under extreme imbalance — high ROC-AUC values are compatible with near-zero fraud recall (Saito and Rehmsmeier, 2015). A third, and perhaps the least-studied limitation, is the treatment of the decision threshold: most

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studies either fix it at the default value of 0.5 or select it by inspecting the test set, both of which bias reported metrics in favour of models with well-separated scores.

This paper addresses the threshold problem directly. The decision threshold τ that converts a model's output probability into a binary fraud/legitimate decision is not a neutral operational parameter: on datasets where fraud scores are concentrated far below 0.5, a fixed $\tau = 0.5$ effectively predicts all transactions as legitimate, destroying recall. Yet no published study systematically compares multiple threshold strategies across multiple model families and multiple fraud datasets under a strict leakage-free protocol.

This paper makes the following contributions:

- C1. Leakage-free joint benchmark.** A reproducible evaluation protocol is established for six model families on two structurally distinct fraud datasets (ULB and BAF), eliminating data leakage at every stage: resampling, threshold selection, and hyperparameter tuning.
- C2. Threshold sensitivity as a primary experimental variable.** Four threshold selection strategies are systematically compared — fixed $\tau = 0.5$, $\max-F_1$, $\max-F_2$, and Precision ≥ 0.5 — each calibrated exclusively on inner validation folds, with quantified impact on both metric values and model rankings.
- C3. Full factorial analysis of imbalance strategies.** Seven class imbalance handling strategies are evaluated in a $7 \times 5 \times 2$ factorial design (strategy $\times$ model $\times$ dataset), identifying which interventions help, harm, or leave performance unchanged.
- C4. Practitioner decision framework.** A decision table maps seven representative deployment scenarios to recommended model–strategy–threshold configurations, providing directly actionable guidance.

The remainder of this paper is structured as follows. Section 2 surveys the relevant literature on fraud detection evaluation, class imbalance, and threshold selection. Section 3 describes the experimental framework. Section 4 presents the experimental results. Section 5 discusses the main findings and the decision framework. Section 6 concludes and outlines directions for future work.

2. Related Work

2.1. Fraud Detection Benchmarks and Evaluation under Class Imbalance

The ULB Credit Card Fraud dataset (Dal Pozzolo, Caelen, Johnson and Bontempi, 2015; Dal Pozzolo, Boracchi, Caelen, Alippi and Bontempi, 2017) has been the de facto benchmark in the field for over a decade, with the important caveat that its features are PCA-anonymised and therefore not interpretable in domain terms. The Bank Account Fraud (BAF) suite (Jesus, Pombal, Alves, Cruz, Saleiro, Ribeiro, Gama and Bizarro, 2022), released at NeurIPS 2022, addresses this gap: it provides one million synthetic but feature-rich bank account applications with semantic variable names, a $\approx 1.1\%$ fraud rate, and five distribution-shifted variants designed to stress-test model robustness under covariate shift, label shift, and subpopulation bias. The two datasets are complementary: ULB is small ($N = 284K$) and privacy-constrained; BAF is large-scale with interpretable features.

A key methodological issue with fraud detection benchmarks is the choice of evaluation metric. Saito and Rehmsmeier (2015) demonstrate that ROC-AUC is insensitive to the class prior and therefore misleading under extreme imbalance: a model with ROC-AUC > 0.95 may still fail to detect most fraud cases. Area under the Precision-Recall curve (PR-AUC) is a more informative measure, as it directly quantifies the trade-off between fraud detection rate (recall) and the burden of false positives (precision) at all possible operating points (Saito and Rehmsmeier, 2015; Dal Pozzolo et al., 2017). The F_β metric, which weights recall β -times more than precision, is widely used as the operational metric under deployment constraints (Ryman-Tubb et al., 2018). The F_2 variant ($\beta = 2$) is particularly relevant for fraud detection, where missing a genuine fraud transaction typically carries a greater cost than investigating a false positive.

2.2. Class Imbalance Handling Strategies

The principal strategies for handling class imbalance in supervised learning fall into three categories: under-sampling (reducing the majority class), oversampling (augmenting the minority class), and cost-sensitive learning (modifying the loss function).

Table 1
Dataset characteristics.

Dataset	N	Fraud (%)	Features	Type	Reference
ULB 2013	284,807	0.17	30	PCA + Amount	Dal Pozzolo et al. (2017)
BAF Base	1,000,000	1.09	30	Semantic	Jesus et al. (2022)

SMOTE (Chawla, Bowyer, Hall and Kegelmeyer, 2002) is the most widely cited oversampling technique: it generates synthetic minority-class examples by interpolating between existing fraud transactions in feature space. Batista, Prati and Monard (2004) extended SMOTE with Tomek Links cleaning (SMOTE+Tomek) and SMOTEENN, which apply selective removal of majority-class borderline examples after oversampling. Comprehensive reviews (He and Garcia, 2009; ?) conclude that no single strategy consistently outperforms others across datasets and model families, underscoring the need for factorial evaluation rather than single-strategy comparisons.

A practical limitation of most comparative studies is that imbalance strategies are evaluated in isolation from the threshold selection step: the strategy is fixed, and the threshold is fixed at 0.5 or chosen informally. This conflation prevents attribution of performance differences to their true source — a threshold that is optimal for the imbalanced distribution may be entirely different from 0.5, independently of whether oversampling was applied.

2.3. Threshold Selection and Evaluation Integrity

The decision threshold is the value above which a model’s output probability is classified as fraud. In the standard scikit-learn interface, the default threshold is 0.5, calibrated for balanced binary classification. Under extreme imbalance, fraud model scores are systematically depressed: the empirically optimal threshold on bank account fraud data lies in the 0.05–0.10 range, far from the 0.5 default. Fixing $\tau = 0.5$ in this regime is equivalent to predicting all transactions as legitimate, producing recall near zero and F_2 near zero.

Luzio, Ponti, Arevalo and Argerich (2024) formally decoupled the threshold selection step from model training in the context of fraud detection, showing that this separation eliminates an important source of evaluation bias. Hayat and Magnier (2025) catalogued data leakage patterns in the fraud detection literature and showed that threshold selection on the test set is among the most common sources of inflated performance claims. Dal Pozzolo et al. (2017) compared threshold strategies on ULB and demonstrated that model rankings can change under different threshold choices, yet most subsequent work has not adopted this finding into standard evaluation practice.

In the tabular deep learning literature, Shwartz-Ziv and Armon (2022) showed that gradient-boosted decision trees (GBDTs) consistently match or outperform tabular transformers on small-to-medium tabular datasets, a finding that motivates the model selection in this paper. However, this comparison was conducted without a systematic threshold sensitivity study, leaving open the question of whether the observed performance differences are robust to the choice of evaluation threshold.

The present paper addresses this gap directly: to the best of our knowledge, no published study provides a joint leakage-free comparison of classical ML models and a tabular transformer on both a small and a large-scale fraud benchmark with threshold selection treated as a primary experimental variable.

3. Experimental Framework

3.1. Datasets

Table 1 summarises the two datasets used in this study.

ULB Credit Card Fraud 2013. European credit card transactions over two days in September 2013. Features V1–V28 are PCA components (privacy-preserving), plus Time and Amount. The dataset contains 492 fraud cases out of 284,807 transactions (0.172%). Because features are PCA-anonymised, domain interpretation is not possible on this dataset.

Bank Account Fraud (BAF) Base. One million synthetic bank account applications generated via the simulation framework of Jesus et al. (2022) (NeurIPS 2022 Datasets Track). The base dataset contains 30 semantic features (25 numerical, 5 categorical: `payment_type`, `employment_status`, `housing_status`, `source`, `device_os`), and a binary label `fraud_bool` with approximately 1.09% fraud prevalence. Unlike ULB, feature names carry domain meaning, enabling interpretability analysis.

3.2. Models

Six model families are evaluated:

- **Logistic Regression (LR)**: linear baseline with elastic-net regularisation (?); serves as a lower-bound reference.
- **Random Forest (RF)**: bagged ensemble of decision trees (?); known for cross-domain robustness via feature subsampling.
- **LightGBM (LGBM)**: gradient-boosted trees with leaf-wise growth and histogram binning (Ke, Meng, Finley, Wang, Chen, Ma, Ye and Liu, 2017); fast, accurate on tabular data.
- **CatBoost (CB)**: gradient boosting with ordered boosting and native categorical handling (Prokhorenkova, Gusev, Vorobev, Dorogush and Gulin, 2018); strong out-of-the-box performance on mixed-type tabular datasets.
- **One-Class SVM (OCSVM)**: anomaly-detection baseline trained only on legitimate transactions (?); included to quantify the value of supervised fraud labels.
- **FT-Transformer**: Feature Tokeniser + Transformer architecture for tabular data (Gorishniy, Rubachev, Khrulkov and Babenko, 2021); represents the most competitive tabular deep learning architecture in the literature.

3.3. Class Imbalance Handling Strategies

Seven strategies are evaluated for all supervised models (LR, RF, LGBM, CB, FT-Transformer):

1. **None**: no resampling (baseline).
2. **RUS**: Random Under-Sampling — randomly removes majority-class examples.
3. **ROS**: Random Over-Sampling — randomly duplicates minority-class examples.
4. **SMOTE** (Chawla et al., 2002): synthetic minority oversampling.
5. **SMOTE+Tomek** (Batista et al., 2004): SMOTE followed by Tomek Links cleaning.
6. **SMOTEENN** (Batista et al., 2004): SMOTE followed by Edited Nearest Neighbour cleaning.
7. **Class Weights**: modifies the model's loss function to penalise minority-class errors proportionally to the inverse class frequency.

OCSVM is excluded from the strategy comparison (it uses no fraud labels). All resampling strategies are applied exclusively within the training partition of each cross-validation fold (see Section 3.5) to prevent resampling leakage. The full design yields $7 \times 5 = 35$ configurations per dataset for supervised models.

3.4. Decision Threshold Strategies

This paper treats the decision threshold as a primary experimental variable rather than a post-hoc operational detail. For each trained model, four threshold strategies are evaluated:

1. **Fixed $\tau = 0.5$** : the naïve default; appropriate only when output scores are well-calibrated around 0.5.
2. **max- F_1** : $\tau^* = \arg \max_{\tau} F_1(\tau)$ on the validation folds (balanced precision-recall trade-off).
3. **max- F_2** : $\tau^* = \arg \max_{\tau} F_2(\tau)$ on the validation folds; primary criterion, weighting recall twice as heavily as precision (appropriate when missed fraud costs more than false alerts).
4. **Precision ≥ 0.5** : the lowest τ such that $\text{Precision}(\tau) \geq 0.5$ on the validation folds; simulates a hard workload constraint (at most one false alert per genuine fraud).

All threshold strategies select τ exclusively on inner validation folds. The median τ across folds is then applied to the held-out test set. Crucially, the threshold study reuses the same trained model and probability scores from the baseline (strategy = None) — *no retraining is performed* — so the entire threshold sensitivity analysis is free of additional computational cost.

3.5. Leakage-Free Evaluation Protocol

A strict leakage-free protocol is enforced at three stages: resampling, preprocessing, and threshold selection.

Outer split. Each dataset is partitioned into an 80% training partition (DEV) and a 20% held-out test set, stratified by class label. The test set is never examined during any selection, fitting, or calibration step.

Inner cross-validation. All model selection and threshold calibration occur within the DEV partition via stratified 5-fold cross-validation. Within each fold: (1) the preprocessor (imputation, standardisation, one-hot encoding) is fitted on the fold's training subset only and applied to the validation subset; (2) resampling is applied only on the fold's training subset; (3) the model is trained on the resampled training subset; (4) the threshold τ is selected on the fold's validation subset using the chosen strategy. The median τ across the five folds is the threshold applied to the test set.

Table 2Baseline performance on ULB 2013 (strategy = None, threshold = max- F_2).

Model	PR-AUC	ROC-AUC	F_1	F_2	Recall	Alert Rate
LR	0.742	0.965	0.749	0.799	0.837	0.21%
RF	0.867	0.959	0.831	0.858	0.878	0.19%
LGBM	0.874	0.975	0.824	0.843	0.857	0.19%
CatBoost	0.885	0.978	0.833	0.853	0.867	0.19%
FT-Trans.	0.856	0.975	0.755	0.807	0.847	0.17%
OCSVM	0.335	0.953	0.415	0.527	0.643	0.36%

Table 3Baseline performance on BAF Base (strategy = None, threshold = max- F_2).

Model	PR-AUC	ROC-AUC	F_1	F_2	Recall	Alert Rate
LR	0.143	0.875	0.187	0.287	0.445	4.14%
RF	0.159	0.878	0.205	0.294	0.415	3.36%
LGBM	0.177	0.899	0.222	0.316	0.439	3.25%
CatBoost	0.180	0.898	0.211	0.314	0.468	3.80%
FT-Trans.	0.180	0.897	0.225	0.316	0.434	3.16%
OCSVM	0.019	0.628	0.045	0.081	0.171	7.32%

Final evaluation. The model is retrained on the full DEV partition with the best hyperparameters. The median threshold from inner cross-validation is applied to the test set without modification. Bootstrap 95% confidence intervals ($B = 1\,000$ iterations) are computed on the test set for PR-AUC and ROC-AUC.

3.6. Hyperparameter Optimisation

All supervised models are tuned using Optuna (Akiba, Sano, Yanase, Ohta and Koyama, 2019) with the Tree-structured Parzen Estimator (TPE) sampler and MedianPruner (50 trials, PR-AUC objective, stratified 5-fold CV for classical models; single 80/20 holdout within DEV for FT-Transformer). The best hyperparameters found under the baseline condition (strategy = None) are reused for all resampling strategy experiments, isolating the effect of the balancing technique from model capacity changes. Hyperparameter search spaces and the optimal configurations per dataset are reported in the supplementary material.

3.7. Evaluation Metrics

PR-AUC (area under the Precision-Recall curve) is the primary summary metric. The F_β -score is the primary operational metric, with $\beta = 2$ for the threshold sensitivity study (recall-weighted) and $\beta = 1$ for secondary analysis. **Recall** (true positive rate) measures the fraction of fraud cases detected. **Alert Rate** is the fraction of all transactions flagged as fraud, quantifying the operational workload imposed on fraud investigators. **Bootstrap 95% CI** bounds provide statistical precision estimates for PR-AUC and ROC-AUC on the held-out test set.

4. Results

4.1. Baseline Model Comparison

Tables 2–?? report performance under the baseline condition (strategy = None, threshold = max- F_2) on both datasets. Figure 1 shows the corresponding PR-AUC bar charts.

On ULB 2013, all supervised models achieve high PR-AUC (> 0.74). CatBoost leads at 0.885, followed by LGBM (0.874), RF (0.867), and FT-Transformer (0.856). Logistic Regression trails substantially (0.742), and OCSVM serves only as a weak anomaly-detection baseline (0.335). The performance gap between GBDTs/FT-Transformer and LR reflects the ability of non-linear models to capture complex interactions even in PCA-anonymised features. This result is consistent with the finding of Shwartz-Ziv and Armon (2022) that tabular transformers are competitive but rarely superior to gradient boosting on small tabular datasets.

Table 4Threshold sensitivity on ULB 2013 — F_2 by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.664	0.772	0.799	0.742
RF	0.830	0.832	0.858	0.772
LGBM	0.803	0.825	0.843	0.726
CatBoost	0.816	0.816	0.853	0.752
FT-Trans.	0.825	0.825	0.820	0.778
OCSVM	0.420	0.498	0.527	0.455

Table 5Threshold sensitivity on BAF Base — F_2 by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.015	0.245	0.287	0.032
RF	0.001	0.263	0.294	0.045
LGBM	0.042	0.260	0.316	0.056
CatBoost	0.037	0.274	0.314	0.058
FT-Trans.	0.035	0.244	0.317	0.070
OCSVM	0.031	0.071	0.081	0.000

Table 6

Threshold sensitivity on BAF Base — Alert Rate (%) by threshold strategy.

Model	Fixed (0.5)	max- F_1	max- F_2	Prec ≥ 0.5
LR	0.02	1.59	4.14	0.06
RF	0.00	1.55	3.36	0.08
LGBM	0.07	1.43	3.25	0.10
CatBoost	0.05	1.59	3.80	0.10
FT-Trans.	0.05	1.14	3.66	0.12
OCSVM	1.03	4.06	7.32	0.00

On BAF Base, the picture changes substantially: all supervised models exhibit PR-AUC near 0.18, reflecting the greater difficulty of this large-scale dataset. Notably, the FT-Transformer matches CatBoost exactly at PR-AUC = 0.180, suggesting that attention-based architectures benefit from increased training data ($N = 1\text{M}$ vs. 284K) and close the gap with GBDTs. LGBM (0.177) and RF (0.159) complete the supervised tier; OCSVM collapses (0.019).

4.2. Threshold Sensitivity Analysis

Tables 4–6 report F_2 and Alert Rate under all four threshold strategies on both datasets.

On ULB 2013, threshold choice has a moderate but meaningful impact. For CatBoost and LGBM, the fixed $\tau = 0.5$ already performs well ($F_2 \approx 0.81$) because predicted scores are sufficiently separated. Logistic Regression benefits most from threshold optimisation: F_2 improves from 0.664 (fixed) to 0.799 (max- F_2), a 20% relative gain. Model rankings are preserved across all four strategies on ULB.

On BAF Base, the impact is an order of magnitude larger. The fixed $\tau = 0.5$ threshold is catastrophic: F_2 drops below 0.04 for every supervised model, and RF produces virtually no fraud detections ($F_2 = 0.001$, Alert Rate = 0.00%). This occurs because fraud model scores on BAF are concentrated in the range 0.04–0.10; the default $\tau = 0.5$ discards virtually all positive predictions. The max- F_2 strategy recovers substantial performance: CatBoost F_2 improves from 0.037 to 0.314, a $\approx 8.5\times$ improvement; the FT-Transformer improves from 0.035 to 0.317, a $9\times$ improvement.

The Alert Rate table (Table 6) reveals the workload trade-off: max- F_2 requires investigators to review 3.2–4.1% of all transactions ($\approx 32\,000$ – $41\,000$ per million), whereas the Precision ≥ 0.5 strategy reduces this to 0.06–0.12% at the cost of substantially lower recall. **The key finding is that threshold selection is not a minor post-hoc detail but a**

Table 7PR-AUC by model $\times$ strategy on ULB 2013. Bold: best result per row.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.742	0.674	0.711	0.721	0.721	0.732	0.709
RF	0.867	0.669	0.873	0.870	0.870	0.853	0.859
LGBM	0.874	0.678	0.727	0.724	0.724	0.724	0.726
CatBoost	0.885	0.617	0.879	0.857	0.857	0.846	0.875
FT-Trans.	0.856	0.650	0.836	0.845	0.845	0.772	0.783

Table 8PR-AUC by model $\times$ strategy on BAF Base. Bold: best result per row.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.143	0.137	0.137	0.135	0.135	0.136	0.122
RF	0.159	0.135	0.149	0.120	0.120	0.124	0.146
LGBM	0.177	0.171	0.181	0.146	0.146	0.158	0.181
CatBoost	0.180	0.178	0.185	0.154	0.154	0.165	0.184
FT-Trans.	0.180	0.159	0.166	0.106	0.106	0.124	0.175

Table 9 F_2 by model $\times$ strategy on BAF Base (max- F_2 threshold). Bold: best result per row.

Model	None	RUS	ROS	SMOTE	SM+T	SMOTEENN	Weights
LR	0.287	0.283	0.283	0.280	0.280	0.282	0.267
RF	0.294	0.288	0.303	0.280	0.280	0.280	0.296
LGBM	0.316	0.318	0.319	0.293	0.293	0.300	0.320
CatBoost	0.314	0.317	0.316	0.303	0.303	0.309	0.317
FT-Trans.	0.316	0.289	0.310	0.255	0.255	0.269	0.301

first-order design decision: on BAF, the choice of threshold strategy changes F_2 by an order of magnitude and reverses the operational classification from “detect nothing” to “detect 43–47% of fraud”.

4.3. Full Factorial Analysis: Imbalance Strategy $\times$ Model

Tables 7–9 present PR-AUC and F_2 under all strategy–model combinations. Figure 2 shows the Δ PR-AUC heatmaps (strategy minus None baseline).

On ULB 2013, the baseline (None) is optimal for LGBM, CatBoost, and FT-Transformer. Random Forest is the only model that benefits from oversampling: ROS raises PR-AUC from 0.867 to 0.873. ROS substantially degrades LGBM ($0.874 \rightarrow 0.727$, -17%), suggesting that random duplication of minority samples disrupts its gradient landscape. SMOTE and SMOTE+Tomek produce *identical* results across all models and both datasets, confirming that Tomek Links cleaning removes zero borderline examples when the minority class is extremely rare — the cleaned set is unchanged from the SMOTE output. This equivalence has a practical implication: on fraud data, the added preprocessing cost of SMOTE+Tomek provides no benefit over plain SMOTE.

On BAF Base, well-tuned GBDTs (LGBM, CatBoost) handle class imbalance natively: no strategy improves their PR-AUC beyond the baseline by more than 0.005. SMOTE degrades the FT-Transformer by 41% on BAF Base (PR-AUC $0.180 \rightarrow 0.106$) compared to only 14% for CatBoost ($0.180 \rightarrow 0.154$), an architecture-specific sensitivity discussed in Section 5.3.

4.4. Confidence Intervals and Computational Cost

Tables 10 and 11 report bootstrap confidence intervals and training cost on BAF Base.

The bootstrap CIs (Table 10) confirm that the FT-Transformer and CatBoost confidence intervals overlap completely (CatBoost: $[0.164, 0.197]$; FT-Transformer: $[0.164, 0.197]$), providing no statistical basis for preferring one over the other on BAF at the current sample size. The CI widths reflect the difficulty of estimating PR-AUC under

Table 1095% Bootstrap CI for PR-AUC and ROC-AUC on BAF Base (strategy = None, $B=1\,000$).

Model	PR-AUC	95% CI	ROC-AUC	95% CI
LR	0.143	[0.131, 0.158]	0.875	[0.867, 0.882]
RF	0.159	[0.145, 0.175]	0.878	[0.870, 0.885]
LGBM	0.177	[0.162, 0.194]	0.898	[0.892, 0.905]
CatBoost	0.180	[0.164, 0.197]	0.898	[0.892, 0.904]
FT-Trans.	0.180	[0.164, 0.197]	0.897	[0.891, 0.904]
OCSVM	0.019	[0.018, 0.021]	0.628	[0.616, 0.639]

Table 11

Computational cost on BAF Base (strategy = None). Tuning includes 50 Optuna trials.

Model	Tuning (s)	Training (s)	Inference (s)
LR	55,070	83.9	0.321
RF	105,313	1,494.0	21.075
LGBM	1,243	16.1	1.387
CatBoost	7,010	36.7	0.249
FT-Trans.	46,550	346.9	2.419
OCSVM	—	919.6	71.579

extreme imbalance: even with $N = 1\text{M}$, the bootstrap distribution has substantial variance because the relevant region of the precision-recall curve is determined by approximately 11 000 fraud cases.

Table 11 highlights the substantial computational asymmetry: LGBM requires only 16 seconds of training time and achieves a PR-AUC of 0.177, while the FT-Transformer requires 347 seconds (GPU-accelerated) for a statistically equivalent result of 0.180. LightGBM's Optuna tuning is also 37× faster than FT-Transformer tuning (1 243 s vs. 46 550 s), primarily because LGBM's tree-based training is CPU-parallelisable without epoch-based overhead.

5. Discussion

5.1. Threshold Selection Dominates Model Selection

The threshold sensitivity results reveal a striking asymmetry between the two datasets. On ULB 2013, model choice is the dominant factor: the spread in PR-AUC across models (0.742–0.885) is large, and the fixed $\tau = 0.5$ default already performs reasonably ($F_2 \approx 0.80$ –0.83 for tree-based models). This is explained by score calibration: on ULB, fraud transactions receive substantially higher probability scores than legitimate ones, so the default threshold of 0.5 is close to optimal.

On BAF Base, the picture inverts: threshold choice is the dominant factor. Model PR-AUC values cluster within a 0.04-point range (0.143–0.180), while the F_2 spread across threshold strategies spans an entire order of magnitude (0.001–0.317). The mechanistic explanation is probability compression: with $\approx 1.1\%$ fraud prevalence and a feature space that provides only moderate discriminative signal, all models assign fraud scores in the range 0.04–0.10. No model's scores approach the 0.5 default; a fixed threshold of 0.5 is equivalent to predicting all transactions as legitimate.

A second observation from the Alert Rate table (Table 6) is the workload trade-off. The max- F_2 strategy, which maximises recall at the cost of precision, requires investigators to review 3.2–4.1% of all transactions; the Precision ≥ 0.5 strategy reduces this to 0.06–0.12% while accepting substantially lower recall. The appropriate strategy is deployment-context dependent: a high-volume automated system can tolerate a higher alert rate, while a team of human investigators with limited capacity would prefer the precision-constrained strategy. In either case, the deployment team must select a threshold explicitly, with awareness of the metric trade-offs — and must never rely on the default 0.5.

Table 12

Decision framework: recommended configuration by deployment scenario. Recommendations are derived from the results in Section 4.

Deployment scenario	Model	Strategy	Threshold
High recall, small dataset	CatBoost	None / ROS	$\max-F_2$
Balanced F_1 , small dataset	CatBoost	None	$\max-F_1$
High recall, large dataset	LGBM	None / Weights	$\max-F_2$
Balanced F_1 , large dataset	CatBoost	ROS	$\max-F_1$
Low investigation budget	CatBoost	None	$\text{Prec} \geq 0.5$
Interpretability required	LGBM	None	$\max-F_2$
Tabular transformer (DL)	FT-Trans.	Class Weights	$\max-F_2$

5.2. No Imbalance Strategy Universally Dominates

The factorial results confirm that no single imbalance handling strategy improves performance across all model families and both datasets. Two concrete findings stand out.

First, **SMOTE and SMOTE+Tomek are empirically equivalent** on both datasets: every cell in the PR-AUC factorial table is identical for the two strategies. This occurs because Tomek Links cleaning, which removes borderline majority-class examples near synthetic minorities, removes zero examples when the minority class is as rare as 0.17%–1.1%: the synthetic SMOTE examples are so sparse relative to the majority class that no borderline pairs exist to clean. Practitioners who use SMOTE+Tomek incur additional preprocessing cost for no empirical gain on data of this sparsity.

Second, **Class Weights is, on balance, the safest default** across model families: it never degrades GBDT or FT-Transformer performance substantially on BAF, and it occasionally improves it (LGBM +0.004, CatBoost +0.004). The principal exception is LGBM on ULB, where Class Weights drops PR-AUC from 0.874 to 0.726 (−17%) — but crucially, every other strategy also degrades LGBM on ULB by a similar margin (ROS: −17%; SMOTE: −17%). This pattern indicates that LGBM on ULB is sensitive to any deviation from the baseline (None) due to dataset-specific characteristics, not a Class Weights-specific failure. In the absence of prior knowledge about strategy effectiveness on the target dataset, Class Weights is the recommended starting point.

On ULB, Random Forest is the only model that benefits from oversampling (ROS +0.6% PR-AUC), while all GBDT models perform best with no intervention. On BAF, no strategy improves LGBM or CatBoost by more than 0.005 PR-AUC, consistent with the finding that modern GBDT implementations handle moderate imbalance natively through built-in tree-weighted loss functions.

5.3. Architecture-Specific Imbalance Sensitivity

The comparison of FT-Transformer and CatBoost under resampling strategies reveals an architecture-specific interaction that has direct implications for model deployment. On BAF Base, SMOTE degrades the FT-Transformer's PR-AUC by 41% (0.180 → 0.106), compared to only 14% for CatBoost (0.180 → 0.154). This asymmetry appears in both PR-AUC and F_2 and is fully reproduced by SMOTE+Tomek (identical to SMOTE, as discussed above).

The most likely explanation is architectural: synthetic minority examples generated by SMOTE introduce distributional shifts in feature space that disrupt gradient-based optimisation more severely than tree-based learning, which is insensitive to the global density of the training distribution and relies only on local split statistics. A full mechanistic account — including FT-Transformer attention diagnostics and cross-domain robustness analysis — is provided in a companion paper currently in preparation. The practical implication is immediate: practitioners applying SMOTE to a neural tabular model should validate against a Class Weights baseline before adopting the strategy, as the expected gain for tree-based models does not transfer to attention-based architectures.

5.4. Decision Framework

Table 12 synthesises the experimental evidence into a decision framework for fraud detection practitioners. Each row maps a deployment scenario to the recommended model, imbalance strategy, and threshold strategy, derived from the experimental results in Section 4.

Several patterns emerge from the framework. CatBoost is the default choice for in-domain deployment due to its consistently top-tier performance on both datasets. When investigation capacity is limited, the Precision ≥ 0.5 threshold is recommended: it ensures at least one genuine fraud per two alerts, keeping investigator workload manageable at

the cost of lower recall. When interpretability is a requirement, LGBM is preferred over CatBoost: its SHAP feature importance ranking is identical to CatBoost's on this task (established in our companion paper), while its tuning cost is $5.6\times$ lower and its training time is $2.3\times$ faster (Table 11). For FT-Transformer deployments, Class Weights is the recommended imbalance strategy — SMOTE-based alternatives degrade performance substantially (Section 5.3).

The framework makes a concrete practical recommendation that is directly actionable without additional experimentation: for any fraud detection deployment on data resembling the BAF regime (large-scale, moderate imbalance, score compression), the first step should be to select the threshold explicitly using a recall-weighted criterion on a held-out validation set. The specific model matters far less than this step: on BAF, the range of PR-AUC across all supervised models is 0.04 points, while the range of F_2 across threshold strategies is 0.31 points.

6. Conclusion

This paper presented a systematic, leakage-free comparison of six model families under seven class imbalance handling strategies and four decision threshold strategies across two structurally distinct financial fraud datasets. The experimental design prevents the three most common evaluation failures in the fraud detection literature: resampling leakage, test-set threshold selection, and reliance on ROC-AUC as the primary metric.

The most practically significant finding is the dominant role of threshold selection. On the large-scale BAF Base dataset, the fixed $\tau = 0.5$ default reduces F_2 to near zero for every model, while the max- F_2 strategy recovers up to a nine-fold improvement, regardless of the model or imbalance strategy used. On the small ULB dataset, model choice matters more, with CatBoost achieving the best PR-AUC (0.885) and gradient-boosted models consistently outperforming the FT-Transformer and Logistic Regression. On the larger BAF dataset, the FT-Transformer closes the gap entirely, matching CatBoost at PR-AUC = 0.180, but at substantially higher computational cost.

No imbalance strategy universally dominates. SMOTE and SMOTE+Tomek produce identical results under extreme imbalance (Tomek removes nothing), and Class Weights is the safest default when no prior validation evidence exists. SMOTE is strongly contraindicated for attention-based architectures: it degrades the FT-Transformer by 41% on BAF, more than three times the degradation observed for CatBoost under the same strategy.

These findings lead to a clear practical recommendation: any fraud detection deployment must include an explicit threshold calibration step on a held-out validation set. Fixing $\tau = 0.5$ without justification is an evaluation failure, not a design choice. The decision framework provided in Table 12 operationalises the experimental evidence into actionable deployment recommendations.

Two limitations bound the scope of these conclusions. First, the hyperparameter optimisation budget (50 Optuna trials) may under-explore the FT-Transformer's search space relative to the classical models, whose smaller spaces are more efficiently covered under TPE sampling. Second, the BAF dataset is synthetically generated; whilst it is calibrated to resemble real bank account application fraud, patterns in live production systems may include additional complexity (label noise, adversarial adaptation, temporal feedback loops) not captured in the benchmark. A mechanistic explanation of *why* gradient-boosted models match or outperform tabular transformers on fraud data — including FT-Transformer attention diagnostics and cross-domain robustness analysis across the five BAF variants — is provided in a companion paper.

CRedit authorship contribution statement

Olavo Caixeiro: Conceptualization, Methodology, Software, Formal analysis, Writing – original draft. **Maryam Abbasi:** Conceptualization, Supervision, Writing – review & editing. **Pedro Martins:** Supervision, Writing – review & editing.

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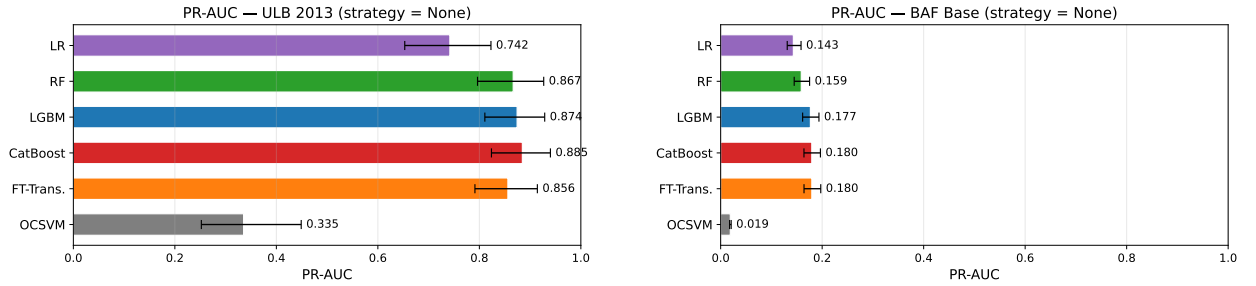


Figure 1: PR-AUC for all models on ULB 2013 (left) and BAF Base (right), baseline condition (strategy = None, max- F_2 threshold).

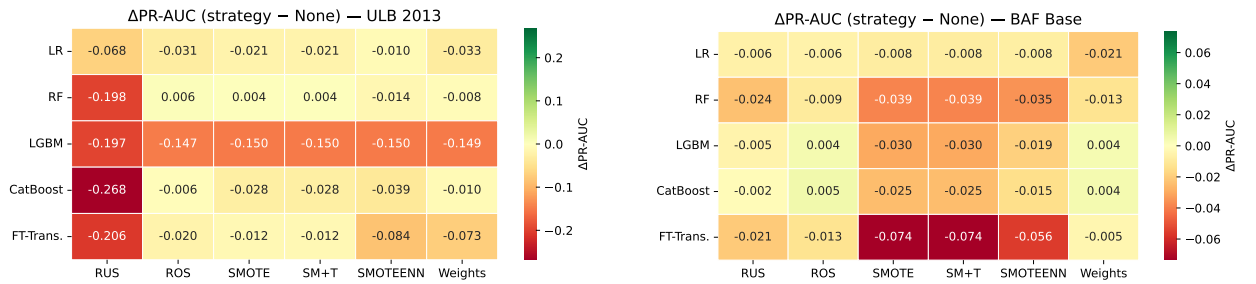


Figure 2: Δ PR-AUC (strategy minus None baseline) heatmaps on ULB 2013 (left) and BAF Base (right). Green cells indicate improvement; red cells indicate degradation.